

## 9.1 Preference Data and Reward Models

Bradley–Terry on a pair. Chance is  $\log 2$ . Best-of- $n$  uses the RM without PPO.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

SFT copies demonstrations. **Preferences** say which of two replies is better when both are fluent. A **reward model (RM)** is a classifier of that comparison. Week 9.2 will use the RM inside RL; Week 9.3 will skip it.

---

### 1. What a reward model is

A human (or a strong teacher model) sees a prompt  $x$  and two completions  $y_w$  (**chosen**, winner) and  $y_l$  (**rejected**, loser). The label is a pairwise order, not a numeric grade. Ties and multi-way ranks exist; this course uses pairs. A **reward model (RM)** is a scalar scorer  $r_\phi(x, y)$  trained so that the chosen reply gets a higher number than the rejected one.

Bradley–Terry turns that comparison into a logistic: the probability that  $y_w$  beats  $y_l$  is  $\sigma(r_\phi(x, y_w) - r_\phi(x, y_l))$ . The loss is  $-\log$  of that probability. Only **differences** enter.  $r_\phi$  is not a grade out of 10, not truth, and not “helpfulness” in the abstract. It is a fit to the raters you hired, on the prompts you wrote.

If the two rewards are equal, the argument of  $\sigma$  is 0 and  $\sigma(0) = 1 / 2$ , so the loss is  $\log 2 \approx 0.693$ . That is chance: the RM has not ordered the pair. Lab 9 will fit this on toy scores.

---

### 2. Why we use it

SFT copies a single answer. Preferences say “this is better than that” without writing a scalar label by hand. It is easier to compare than to write, which is why we collect pairs, not essays.

A base LM that already speaks still needs a way to rank two fluent replies. The RM is that ranker. At test time you can rank  $n$  samples from the policy by  $r_\phi$  (**best-of- $n$** ) without any RL. Best-of- $n$  already uses this RM and is a competitive baseline (AlpacaFarm). It is not PPO. You sample  $n$  from SFT and return the RM’s favorite. PPO (next note) trains a new  $\pi_\theta$ .

Collection is the bottleneck: instructions must be clear, raters disagree, and the pair set can encode rater bias. If both  $y_w$  and  $y_l$  are wrong, the RM still learns a relative order among mistakes. Held-out preference **accuracy**—how often  $r_\phi(x, y_w) > r_\phi(x, y_l)$  on pairs the RM did not train on—is the number to report. Do not report only training loss.

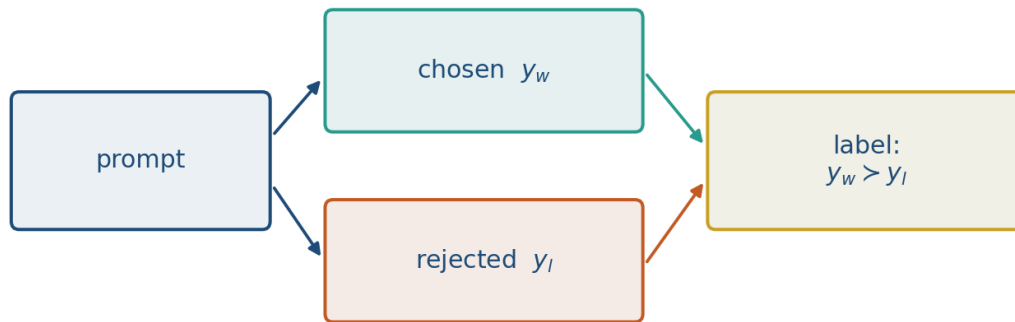
---

### 3. Architecture

Initialize from the SFT policy (or its encoder). The RM already speaks the same dialect as the completions it must score. A raw pretrained LM does not format like an assistant, so its scalar head starts on the wrong geometry.

Usually you keep the base LM and add a **scalar head** on the last token, or you train a small classifier that reads the completion. There is no PPO loop in this note. Best-of- $n$  is the serve-time architecture: sample  $n$ , pick  $\arg \max r$ .

One preference row is a comparison, not a score out of 100



A reward model is a classifier of “better completion,” trained on pairs



 Three  $\Delta$  values and the logistic loss

$\Delta$  values and the logistic loss" src="graphics/9.1-preference-rewards/rm-numeric.png" />

The picture is a pair in, a difference of scalars, a logistic loss. Lab 9 is a tiny head on toy scores so you can print  $\Delta$  and  $\sigma(\Delta)$  without an LLM.

---

### 4. How it works, step by step

1. **Collect pairs.** Prompt  $x$ , chosen  $y_w$ , rejected  $y_l$ . Clear instructions to raters; expect disagreement.

2. **Initialize**  $r_\phi$  from SFT. Score each completion with a scalar.
3. **Train** the Bradley–Terry / logistic loss below. Chance loss is  $\log 2$ . Large positive  $\Delta \rightarrow$  loss near 0; large negative  $\Delta$  (chosen scored **below** rejected)  $\rightarrow$  loss near  $\infty$ .
4. **Check held-out accuracy**, not only train loss. The RM is a pairwise logistic classifier, not a calibrated “helpfulness out of 10.”
5. **Optional decode**: best-of- $n$ . Sample  $n = 8$  completions from  $\pi_{\text{SFT}}$ , score each with  $r_\phi$ , return  $\arg \max_i r_\phi(x, y_i)$ . Extra cost: 8 policy forwards plus 8 RM forwards (the RM is usually cheaper). No PPO. The distribution is “SFT, then pick the RM’s favorite,” not a new  $\pi_\theta$ . If two samples share the top score, freeze a tie-break (lower id, or a second sample) so a demo is reproducible; the RM failed to separate them.

Reward hacking waits in the next notes: the policy learns to please  $r_\phi$  with verbose, sycophantic, or empty-safe text. That is why RLHF adds a KL penalty to the SFT model, and why you keep a held-out preference set.

---

## 5. Mathematical formulas

Bradley–Terry probability and RM loss:

$$p(y_w > y_l | x) = \sigma(r_\phi(x, y_w) - r_\phi(x, y_l)),$$

$$\mathcal{L}_{\text{RM}} = -\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l)).$$

$$\sigma(\Delta) = (1 + e^{-\Delta})^{-1}. \text{ If } r_w = r_l, \Delta = 0, \sigma(0) = 1/2, \mathcal{L}_{\text{RM}} = \log 2 \approx 0.693 \text{ nats per pair.}$$

Worked numbers live in the example below:  $\Delta = 2$  gives  $\sigma \approx 0.881$  and loss  $\approx 0.127$ ; a flipped pair  $\Delta = -2$  gives loss  $\approx 2.13$ . Compute  $\log 2$  once as a class so “0.693 at init” is a number people can check in Lab 9, not a magic constant.

---

## 6. Positive points and negative points

### Positive.

- Pairwise data is easier to collect than absolute scores or full essays.
- The same RM already gives a strong decode baseline: best-of- $n$ , no PPO.
- Init from SFT puts the scalar head on assistant-shaped text.
- Held-out preference accuracy is a number you can rerun after a policy update.

### Negative.

- Reward hacking: a later policy can please  $r_\phi$  with verbose, sycophantic, or empty-safe text.
- RM error becomes policy error. A relative order among two wrong answers is still an order.

- Best-of- $n$  spends compute at decode ( $n$  policy forwards plus  $n$  RM forwards).
- $r_\phi$  is not truth or safety in the abstract; it is a fit to your raters and prompts.
- Reporting only training loss hides a held-out accuracy that never left chance.

**When not to.** Do not call best-of- $n$  “RLHF.” It uses the RM; it does not train a policy with PPO. If you have no preference pairs, you do not have an RM—you have SFT.

---

## 7. Teaching this note

About **30–35 minutes** at the board: one pair, write  $\sigma(\Delta r)$ , compute  $\sigma(0)$ , then best-of- $n$  cost. Play Karpathy’s **RLHF / comparison** appendix **21:05–25:43**. Lab 9 section 1 is the toy logistic RM.

Compute  $\log 2$  once as a class so “0.693 at init” is a number people can check in Lab 9, not a magic constant.

---

## 8. Worked example

Rewards  $r_w = 2.0$ ,  $r_l = 0.0$ . Difference 2.  $\sigma(2) = 1 / (1 + e^{-2}) \approx 0.881$ . Loss  $-\log 0.881 \approx 0.127$ .

If  $r_w = r_l = 1.7$  always,  $\Delta = 0$ ,  $\sigma(0) = 1 / 2$ ,  $\mathcal{L} = -\log(1 / 2) = \log 2 \approx 0.693$  nats per pair. A random RM sits here.

Best-of- $n$ : sample  $n = 8$  completions from  $\pi_{\text{SFT}}$ , score each with  $r_\phi$ , return  $\arg \max_i r_\phi(x, y_i)$ . Extra cost: 8 policy forwards plus 8 RM forwards (the RM is usually cheaper). No PPO. The distribution is “SFT, then pick the RM’s favorite,” not a new  $\pi_\theta$ .

Init from SFT: the RM already speaks the same dialect as the completions it must score. A raw pretrained LM does not format like an assistant, so its scalar head starts on the wrong geometry.

If  $r_w = 0$ ,  $r_l = 2$ ,  $\Delta = -2$ ,  $\sigma(-2) \approx 0.119$ , loss  $-\log 0.119 \approx 2.13$ . The RM is **wrong** on that pair; that is a large gradient. Lab 9 should print both the easy pair and a flipped pair.

---

## 9. Where students get stuck

- Treating  $r_\phi$  as a grade out of 10. Only **differences** enter the loss.
  - Forgetting  $\sigma(0) = 1 / 2$  when the two rewards match.
  - Calling best-of- $n$  “RLHF.” It uses the RM; it does not train a policy with PPO.
- 

## 10. Video

[Karpathy — Intro to Large Language Models](#). Play the **RLHF / preference** section **21:05–25:43** (comparisons, labeling, RLHF). Pause on “it is easier to compare than to write”: that is why we collect

pairs, not essays. PPO details wait for note **9.2**.

---

## 11. Practice

1. If  $r(x, y_w) = r(x, y_l)$  always, what is  $\mathcal{L}_{\text{RM}}$ ?
2. Why initialize the RM from SFT instead of from a raw pretrained LM?
3. Best-of- $n$  uses the RM without PPO. What extra cost did you pay at inference?
4. Compute  $\sigma(r_w - r_l)$  and the RM loss for  $r_w = 1, r_l = -1$ . ( $\sigma(2) \approx 0.881$ .)
5. You score  $n = 4$  samples. Two share the top RM score. How do you break the tie in a reproducible demo, and what did the RM fail to do?